

outstanding quality and design.

Following are the main challenging parameters in mines, where power electronics finds potential applications:

- Altitude
- Dust
- Pollution
- Moisture
- Hazardous environments
- Ventilation

Power-electronics-based devices in mining should have the following features:

- High overload design
- Maximum motor care and low harmonics alternatives
- Should operate even at 50°C without power failure
- Be totally sealed (no dust filters) in harsh environments
- Rotating assets like motors should be protected with latest hardware and software

PMC - OLTC (on-load tap changer) is a unique, master-slave motor control system that allows synchronisation of multiple drives and motors without an encoder. The result is smooth, powerful and fast response with least maintenance and supervision. As a result, each motor provides the same torque and homogeneous aging phenomena.

Some programmable logic controllers (PLCs), power communication tools and solar photovoltaic facilities make mining activities more efficient and smarter. Photovoltaic facilities reduce energy dependency in mines to a great extent.

An important objective in mining is to minimise the cost per ton of the materials moved, to achieve higher profitability. Replacing the traditional AC distribution system with a DC one can significantly improve the length of trailing cables.

Further, renewable energy sources and energy storage devices, like batteries, can bring stability and energy efficiency in mines. Medium-voltage DC microgrids and mobile mining also significantly add to the automation of mining process.

AC power has the capacity to step up and step down voltages by means of transformers. When delivering power over long distances, high-

voltage DC lines may have cost advantages over AC transmission lines of the same capacity.

Above a certain distance (600km to 800km), high-voltage DC is cheaper than high-voltage AC. Also, high-voltage AC has an extended transmission capacity due to lack of reactive power. Lower losses and smaller footprints of high-voltage DC make it environmental-friendly. Power delivery to an excavator via a flexible trailing cable may also benefit from DC option. A DC system is also preferred in remote mine-control activities.

Power electronics giving more life to mining

There are different types of mines in the world, some of which are given below.

- Ground mines
- Surface mines
- High-wall mines
- Under sea mines

Equipment used in such mines include the following:

- Mine load/power centres
- Switchgears
- Motor control centres
- Long-wall controls
- Distribution boxes
- Sub-stations
- Capacitor boxes
- Diesel generator sets
- Leaky feeder radio systems
- Vehicle and personal tracking systems

Power centres convey the required power to the corresponding terminal.

Switchgears protect high-voltage circuits in underground and surface mines.

Motor-control centres control the motors used for equipment such as conveyors, pumps and fans. Motors used in mines can range up to thousands of horsepower, with low, medium and high voltages.

Long-wall mining is an effective method of recovering coal and other minerals from deep underground mines. Here, separate long-wall motor controls are required to operate the motors.

Distribution boxes are used to protect low-voltage circuits. These must be enclosed either in explosion-proof

or non-explosion-proof enclosures.

Capacitor banks improve the power factor of an electrical distribution system in mines.

Power electronics have provided a redundant system using VS65 medium-voltage soft starter and XMV 660 medium-voltage variable speed drive to Tizappa mining plant, adding safety, reliability and efficiency to the mine ventilation system. It consists of 460kW motor coupled with 416kV drive to run the fan in an uninterrupted manner.

Intermountain Electronics is a leading company engaged in supplying power electronics in mining situations, particularly in underground and surface mines, and tunnelling environments. It builds new electrical equipment, and is involved in the reconditioning and repairing of existing equipment. These equipment are user-friendly, and meet industry standards and Mine Safety and Health Administration (MSHA) regulations.

Central Mine Planning and Design Institute (electronic division) in Ranchi is actively associated with the identification and implementation of electronics and control systems for underground as well as open-cast projects. It provides voice and data communication as a part of CoalNet project. It also provides various gas detectors and monitors.

Finally

In short, power electronics can play a great role in the modernisation of the mining industry. Companies such as ABB, Siemens, Havells and Intermountain Electronics are coming out with newer and better electronic products to increase the productivity of mines.

Remote process control, renewable energy systems, the Internet of Things (IoT), Big Data, artificial intelligence (AI) and machine learning are sectors where great attention needs to be paid to make power electronic systems more efficient and reliable. Once wrinkles in the power electronics sector are ironed out, it will be possible to make power electronic instruments more efficient and optimal for mining operations. **EFY**